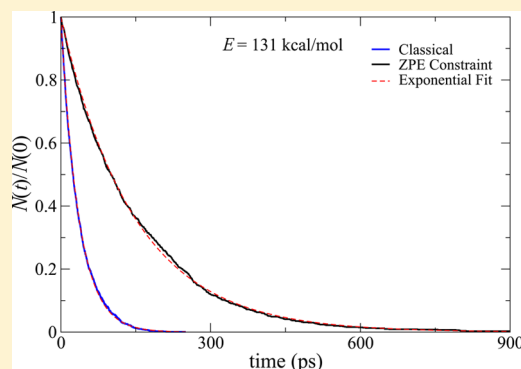


Zero-Point Energy Constraint for Unimolecular Dissociation Reactions. Giving Trajectories Multiple Chances To Dissociate Correctly

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ABSTRACT: A zero-point energy (ZPE) constraint model is proposed for classical trajectory simulations of unimolecular decomposition and applied to $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ decomposition. With this model trajectories are not allowed to dissociate unless they have ZPE in the CH_3 product. If not, they are returned to the CH_4^* region of phase space and, if necessary, given additional opportunities to dissociate with ZPE. The lifetime for dissociation of an individual trajectory is the time it takes to dissociate with ZPE in CH_3 , including multiple possible returns to CH_4^* . With this ZPE constraint the dissociation of CH_4^* is exponential in time as expected for intrinsic RRKM dynamics and the resulting rate constant is in good agreement with the harmonic quantum value of RRKM theory. In contrast, a model that discards trajectories without ZPE in the reaction products gives a $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ rate constant that agrees with the classical and not quantum RRKM value. The rate constant for the purely classical simulation indicates that anharmonicity may be important and the rate constant from the ZPE constrained classical trajectory simulation may not represent the complete anharmonicity of the RRKM quantum dynamics. The ZPE constraint model proposed here is compared with previous models for restricting ZPE flow in intramolecular dynamics, and connecting product and reactant/product quantum energy levels in chemical dynamics simulations.



I. INTRODUCTION

Classical trajectory simulations are important for studying unimolecular reaction dynamics.^{1–6} The fundamental assumption of microcanonical unimolecular rate theory (i.e., RRKM theory)⁶ may be tested by preparing an initial microcanonical ensemble of states in the classical phase space and determine whether the unimolecular decomposition is exponential with the microcanonical rate constant.^{1–3,6} If an initial nonrandom distribution of states is prepared, e.g., by a chemical reaction such as $\text{H} + \text{C}_2\text{H}_4 \rightarrow \text{C}_2\text{H}_5$,⁷ intramolecular vibrational energy redistribution (IVR)⁸ and unimolecular decomposition from this nonrandomly prepared state may be studied.

A shortcoming of classical dynamics is that it does not have zero-point energy (ZPE) constraints and, for statistical RRKM unimolecular decomposition, the reaction barrier is the classical barrier and not the quantum barrier with ZPE.⁹ Thus, when an initial classical microcanonical ensemble is simulated, the comparison for the classical dynamics is with classical, and not quantum, RRKM theory. For an initial nonrandom distribution of unimolecular reactant states, the classical dynamics may give results in good agreement with experiment and quantum dynamics before there is substantial classical IVR and ZPE flow.^{10–12} At short time, classical dynamics may duplicate the vibrational adiabaticity of quantum dynamics.

Models have been proposed and used for addressing the unphysical flow of ZPE in classical chemical dynamics simulations.^{13–21} These unphysical ZPE dynamics can

artificially enhance the rate of intramolecular vibrational energy redistribution (IVR)²² and also lead to the formation of reaction products without ZPE.⁹ For bimolecular reactions, trajectories which do not have ZPE in the reaction products are often discarded.^{17–19} This may be either a soft or hard ZPE constraint,^{20,21} where the former discards trajectories if the sum of the two products' vibrational energies is less than the sum of their ZPEs, whereas the latter discards trajectories if either product has less vibrational energy than its ZPE.

The study reported here was motivated by previous simulations for the $\text{OH}^-(\text{H}_2\text{O}) + \text{CH}_3\text{I}$ reaction.^{23,24} The $\text{S}_{\text{N}}2$ channel, forming $\text{CH}_3\text{OH} + \text{I}^-$, is highly exothermic and the products are formed with ZPE. On the contrary, the proton transfer channel, forming $\text{H}_2\text{O} + \text{CH}_2\text{I}^-$, is endothermic and except at high collision energies most of the trajectories form products without ZPE. For each of these two channels the reactant H_2O molecule may be attached or unattached to the products. For direct proton transfer a soft or hard ZPE constraint, as described above, may be appropriate. However, the majority of the proton transfer is indirect, with transient complex formation, and the applicability of the above constraints is somewhat uncertain for $\text{OH}^-(\text{H}_2\text{O}) + \text{CH}_3\text{I}$ proton transfer mediated by a prereaction complex.

Received: November 19, 2015

Revised: January 5, 2016



Table 1. Properties of the Variational Transition States^a

E_{total}^b	classical			quantum		
	$r^\ddagger$	$V(r^\ddagger)$	$\nu^\ddagger$	$r^\ddagger$	$V_q(r^\ddagger)$	$\nu^\ddagger$
131	2.87	103.73	549.5, 252.3(2)	3.24	96.33	485.8, 157.1(2)
133	2.83	103.36	557.4, 264.1(2)	3.12	95.73	507.0, 188.7(2)

^a $r^\ddagger$ is the H---CH₃ distance in angstrom and $\nu^\ddagger$ the CH₃ umbrella and degenerate H---CH₃ bending frequencies in cm⁻¹ at the variational TS. $V(r^\ddagger)$ is the classical potential at the variational TS and $V_q(r^\ddagger)$ the quantum analog that equals $V_a(r^\ddagger)$ in eq 1 in kcal/mol at the variational TS minus the CH₄ ZPE. ^bEnergy is in kcal/mol.

Previous work^{25,26} indicates that quasiclassical trajectory (QCT) simulations give an accurate rate constant for forming OH⁻(H₂O) + CH₃I prereaction complexes. For the OH⁻ + CH₃I reaction which occurs via exothermic S_N2 and proton transfer pathways, both with prereaction complexes, QCT simulations give temperature dependent reaction rate constants in excellent agreement with experiment.²⁵ Also, a detailed study has shown QCT capture theory often gives rate constants for complex formation in excellent agreement with those obtained by coupled-channel quantum capture theory.²⁶ Thus, for bimolecular reactions which occur via prereaction complexes, a model which discards trajectories without ZPE in the products seems inappropriate. A more complete model may be one that sends trajectories, without ZPE in the reaction products, back to the strong interaction region by reversing the momenta of the products' atoms, giving the trajectories multiple opportunities to dissociate correctly. This is the model considered here. The reactive system considered is CH₄* → H + CH₃ dissociation and details of the simulation model are described in the next section.

II. SIMULATION METHODOLOGY

The chemical dynamics simulations reported here are for CH₄* → H + CH₃ dissociation using the DHS analytic potential energy surface (PES) developed previously.^{27,28} There is another PES for the CH₄ system;^{29,30} however, use of the DHS PES is appropriate for the model simulations reported here. The classical unimolecular dynamics for CH₄ on the DHS PES are ergodic and an initial classical microcanonical ensemble dissociates exponentially with a rate constant in agreement with RRKM theory.³¹ An initial quantum microcanonical ensemble in the classical phase undergoes rapid IVR, forming a classical microcanonical ensemble, which dissociates with classical ergodic dynamics.³¹ In the work presented here quasiclassical microcanonical ensembles³² are prepared for CH₄* on the DHS PES and CH₄* dissociation is simulated at two total vibrational energies E_{total} of 131 and 133 kcal/mol. The total angular momentum J is zero. For one simulation protocol, the dynamics are purely classical and the calculated rate constants versus energy are the classical microcanonical, RRKM values. For the other, a product ZPE constraint is imposed, so that if CH₃ does not contain ZPE at a H---CH₃ separation of 6 Å, the momenta of the H and CH₃ atoms are reversed and the reactive system is sent back to the vibrationally excited CH₄* phase space. The total lifetime of a vibrationally excited CH₄* molecule is then taken as the total time it takes to form products with ZPE in CH₃, including any additional time after revisiting the CH₄* phase space. Except at the high energy classical limit,⁹ the resulting average lifetime of the CH₄* molecules will be longer than the classical value. There are two important issues to consider for this simulation: i.e., does this average lifetime agree with the quantum RRKM value, and with

this ZPE constraint does the dissociation of the CH₄* molecules remain exponential as described by RRKM theory?

III. VARIATIONAL RRKM CALCULATIONS

Harmonic RRKM calculations were performed to compare the classical and quantum RRKM rate constants with those found from the classical and ZPE constrained trajectory simulations. There is not a saddle point for the CH₄ → H + CH₃ potential energy curve; i.e., H + CH₃ association is purely attractive, and variational microcanonical transition state theory (TST)^{33,34} (i.e., RRKM) was used to determine the variational TS structures and RRKM rate constants. The variational TS is located at the minimum in the sum of states $N^\ddagger(E)$ along the reaction path. This minimum arises from a decrease in the transitional mode vibrational frequencies (i.e., the two H---CH₃ bends) and an increase in the potential energy in going from the CH₄ potential energy minimum to the H + CH₃ products.³⁵

Both the harmonic vibrator model,^{36,37} based on the reaction path Hamiltonian,³⁸ and the flexible transition state theory (FTST) model³⁴ have been used to calculate the dissociation and association rate constants for CH₄ ↔ H + CH₃. In addition, the variable reaction coordinate transition state theory (VRC-TST) model^{39–41} has also been used. These three models give similar rate constants for CH₄ ↔ H + CH₃.^{28,39,42,43} To illustrate, for H + CH₃ association at 300 K the vibrator, FTST, and VRC-TST models give canonical variational transition state theory (CVTST) rate constants of 1.43×10^{11} , 1.55×10^{11} , and 1.43×10^{11} L mol⁻¹ s⁻¹, respectively, for DHS PES used here. At 1000 K these respective rate constants are 1.83×10^{11} , 2.17×10^{11} , 1.88×10^{11} L mol⁻¹ s⁻¹. These rate constants are also in excellent agreement with classical trajectory values calculated using the DHS PES.²⁸

For the work presented here the harmonic vibrator model was used to calculate CH₄* → H + CH₃ harmonic RRKM rate constants. With this model the reaction path is identified by the intrinsic reaction coordinate (IRC),⁴⁴ which gives the geometry of the reactive system and the classical potential $V(r^\ddagger)$ along the reaction path. The reaction path Hamiltonian is used to calculate the $3N - 7$ harmonic vibrational frequencies orthogonal to the reaction path to yield the quantum harmonic vibrationally adiabatic potential

$$V_a(r^\ddagger) = V(r^\ddagger) + \sum_{i=1}^{3N-7} h\nu_i(r^\ddagger)/2 \quad (1)$$

The harmonic vibrational frequencies versus $r^\ddagger$ with the classical $V(r^\ddagger)$ and quantum $V_a(r^\ddagger)$ are used to calculate the classical and quantum variational TSs and the corresponding classical and quantum harmonic RRKM rate constants. Properties of the variational transition states are given in Table 1 for the calculations reported here.

IV. RESULTS AND DISCUSSION

For the trajectory initial conditions CH_4^* is randomly excited from a quasiclassical microcanonical ensemble³² with total energies of 131.00 and 133.00 kcal/mol, which includes the CH_4 ZPE of 29.17 kcal/mol. The classical dissociation energy is 109.46 kcal/mol for the PES used here. Therefore, with the CH_3 ZPE of 18.60 kcal/mol, the quantum threshold for forming $\text{H} + \text{CH}_3$ is 128.06 kcal/mol. The E_{total} values of 131.00 and 133.00 kcal/mol, used for the simulations are 3 and 5 kcal/mol higher than the quantum threshold.

A total of 1000 trajectories were calculated for each E_{total} and each trajectory was integrated until dissociation occurred. A distance of 6 Å was selected between CH_3 and H as the criterion for trajectory termination. A distance of 5 Å was not sufficient to identify dissociation, because some trajectories that reached this $\text{H} + \text{CH}_3$ separation returned to CH_4^* .

A. Trajectory Simulations. The ratio of the number of CH_4^* molecules remaining at time t and the initial number of CH_4^* molecules, i.e., $N(t)/N(0)$, is plotted versus time in Figures 1 and 2 for the simulations with E_{total} of 131 and 133

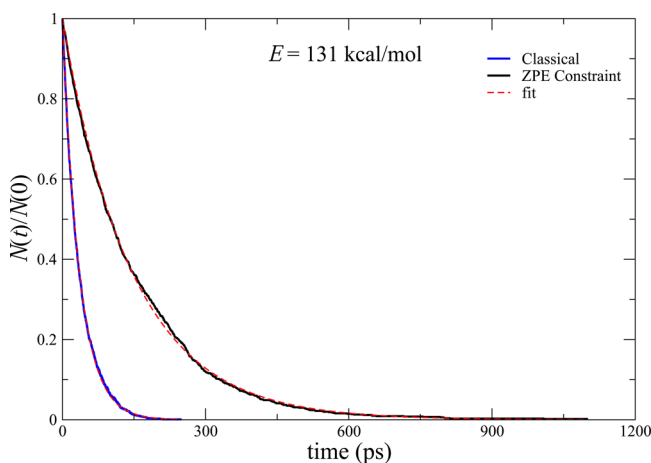


Figure 1. $N(t)/N(0)$ plots versus time for ZPE constrained (black, solid) and classical (blue, solid) trajectory calculations for $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ dissociation at E_{total} 131 kcal/mol. A total of 1000 trajectories are calculated, and all trajectories lead to the dissociation. Also included are the fit with $\exp(-kt)$ (red dashed) to each curve.

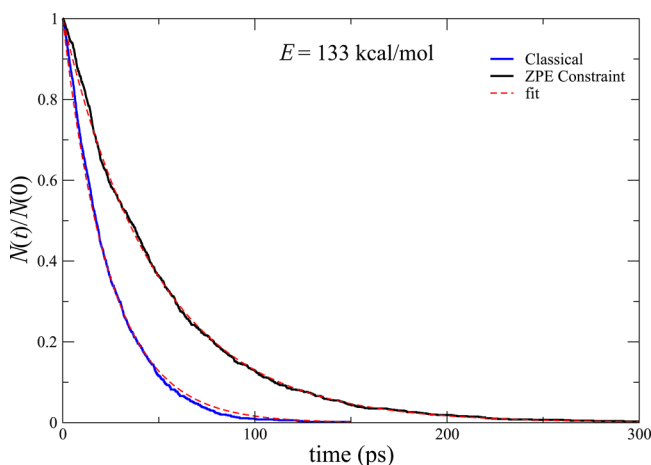


Figure 2. Same as Figure 1 but for E_{total} of 133 kcal/mol.

kcal/mol, respectively. In each figure two $N(t)/N(0)$ plots are presented, i.e., for the classical and ZPE constrained simulations. For both simulations the $N(t)/N(0)$ plots are exponential as expected for intrinsic RRKM behavior. The exponentials provide excellent fits to the trajectory results, with the fitted rate constants listed in Table 2. The classical rate

Table 2. Comparison of Trajectory and Harmonic RRKM Classical and Quantum Rate Constants for $\text{CH}_4 \rightarrow \text{H} + \text{CH}_3$ Dissociation^a

E_{total}^c	Trajectory ^b		RRKM	
	k_c	k_q	k_c	k_q
131	0.0281	0.00684	0.124	0.00728
133	0.0412	0.0204	0.195	0.0141

^aThe classical and quantum rate constants are k_c and k_q , respectively, and in units of ps^{-1} . ^bThe trajectory rate constants k_c and k_q are those for the exponential fits to the simulation $N(t)/N(0)$ distributions, Figures 1 and 2, for E_{total} of 131 and 133 kcal/mol, respectively. ^c E_{total} is in kcal/mol.

constants are 4 and 2 times larger than the ZPE constrained rate constants for E_{total} of 131 and 133 kcal/mol, respectively. The decrease in this ratio with increase in E_{total} is expected from the classical-quantum correspondence.⁹

The relative translational energy distributions of the $\text{H} + \text{CH}_3$ products for the classical and ZPE constrained simulations are compared in Figures 3 and 4 for the 131 and 133 kcal/mol

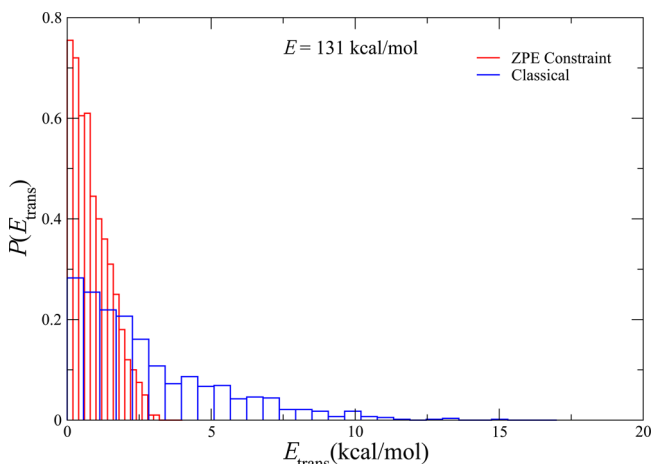


Figure 3. Histogram plots of relative translational energy distribution probability versus relative translational energy for E_{total} of 131 kcal/mol. Red and blue blocks are from the ZPE constrained and classical simulations, respectively. The energy range for each red block is 0.20 and 0.57 kcal/mol for each blue block.

simulations, respectively. The energies available to the products for these two respective simulations are 21.5 and 23.5 kcal/mol and large for the classical simulations, but 2.9 and 4.9 kcal/mol and much smaller for the ZPE constrained simulations. Thus, this distribution for the classical simulation is much broader than the narrow distribution for the ZPE constrained simulation.

B. Comparison with RRKM Theory. The classical and quantum harmonic RRKM rate constants, calculated as described in section III, are listed in Table 2. For the classical simulations the ratio of the trajectory and harmonic RRKM rate constants, i.e., $k_{\text{traj}}(E)/k_{\text{RRKM}}(E)$, is the anharmonic correction

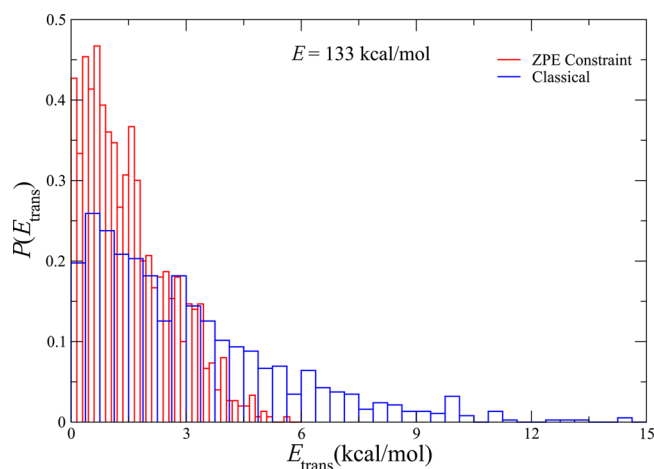


Figure 4. Same as Figure 3, but for E_{total} of 133 kcal/mol. The energy range for each red block is 0.15 and 0.37 kcal/mol for each blue block.

$f_{\text{anh}}(E)$ to the harmonic classical RRKM rate constants.⁴⁵ This correction is 0.23 and 0.21, respectively, for E_{total} of 131 and 133 kcal/mol, and similar to the corrections found previously of 0.40–0.21 for E_{total} increasing from 120 to 150 kcal/mol³¹ and 0.42 for $E_{\text{total}} = 127$ kcal/mol.²⁸ The anharmonic correction factor $f_{\text{anh}}(E)$ is a ratio of the anharmonic corrections for the transition state sum of states and reactant density of states, i.e., $f_{\text{anh}}(E) = f_{\text{anh},N^{\ddagger}}(E)/f_{\text{anh},\rho}(E)$. For the simulations reported here the vibrational energy of the transition state is less than the vibrational energy E_{total} of CH_4^* and, thus, the anharmonic correction $f_{\text{anh},\rho}(E) = \rho_{\text{anh}}(E)/\rho_{\text{h}}(E)$ is expected to be more important than $f_{\text{anh},N^{\ddagger}}(E)$. As determined by three research groups,^{46–48} $f_{\text{anh},\rho}(E)$ is 2 at the CH_4 dissociation energy of ~ 110 kcal/mol, giving $f_{\text{anh}}(E) = 0.5$ at this energy if $f_{\text{anh},N^{\ddagger}}(E)$ is assumed to be unity. Such a value is consistent with the value of 0.4 above for E_{total} of 120–130 kcal/mol. The $f_{\text{anh},\rho}(E)$ of 2 for CH_4^* at the dissociation threshold is consistent with Morse oscillator⁴⁹ and bend–stretch coupling anharmonic corrections.^{50,51}

If the ZPE constrained rate constant is identified as the anharmonic quantum RRKM rate constant, it may be compared with the harmonic quantum RRKM rate constant to identify the anharmonic correction $f_{\text{anh}}(E)$ to the harmonic quantum RRKM rate constant. Making this comparison, one sees that the ZPE constrained trajectory rate constant, determined from the anharmonic PES, is similar to the harmonic quantum RRKM rate constant, with the largest difference of 40% at 133 kcal/mol. Thus, in contrast to the substantial anharmonic correction to the classical RRKM calculation, there is not a significant anharmonic correction between the ZPE constrained anharmonic trajectory and the harmonic quantum RRKM rate constants. More work needs to be done to understand the lack of an anharmonic correction for these latter rate constants in comparison to the classical rate constants. For example, the correspondence between the anharmonic corrections for classical and quantum RRKM rate constants is uncertain. Also, additional analysis is needed to determine the relationship between the ZPE constrained trajectory unimolecular rate constant and the anharmonic quantum RRKM rate constant. If there is not a TS anharmonic correction for the latter, its anharmonic correction will include an anharmonic vibrationally adiabatic reaction path potential energy curve and an anharmonic density of states $\rho_{\text{anh}}(E)$ for CH_4^* .

C. Comparison with Other ZPE Constraint Models. It is of interest to consider other proposed ZPE constraint models. For bimolecular^{17–21} and also unimolecular⁵² reactions, an extensively used model is to discard trajectories which do not form products without ZPE. For $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ dissociation considered here, implementation of this model discards trajectories that do not yield products with ZPE in the CH_3 vibrational modes. The resulting $N(t)/N(0)$ versus t for CH_4^* dissociation is shown in Figure 5a. The calculations are

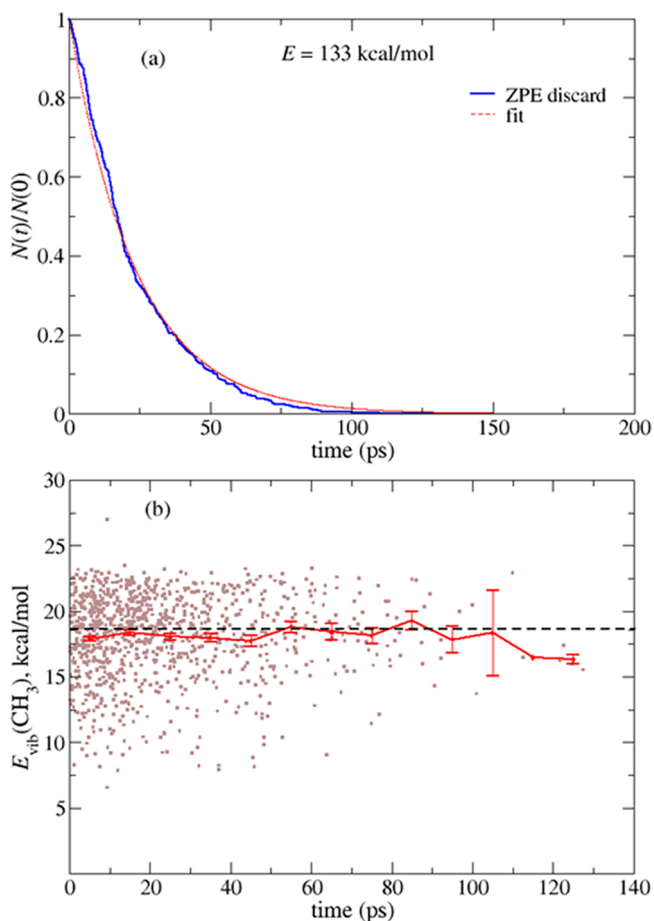


Figure 5. Simulation of $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ dissociation at E_{total} of 133 kcal/mol: (a) $N(t)/N(0)$ resulting from discarding $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ trajectories that do not have ZPE in CH_3 . The rate constant for the exponential fit is 0.43 ps^{-1} . Approximately 50% of the trajectories are discarded. (b) Scatter plot of the CH_3 vibrational energy versus dissociation time for all of the $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ trajectories. The average CH_3 vibrational energy for each 10 ps interval is identified by the solid symbols, connected by straight lines. The 18.6 kcal/mol ZPE of CH_3 is denoted by the dashed line.

for a CH_4 vibrational energy of 133 kcal/mol, 5 kcal/mol higher than the quantum mechanical threshold. Approximately 50% of the trajectories are rejected. The $N(t)/N(0)$ plot is exponential and its rate constant of 0.043 ps^{-1} is nearly the same as that of 0.041 ps^{-1} for the classical dynamics shown in Figure 2. Thus, discarding trajectories which do not have ZPE in the CH_3 product results in a rate constant that agrees with the classical, and not the quantum, RRKM value.

Because the classical dynamics agrees with RRKM theory, it is expected that discarding trajectories without product ZPE results in a rate constant in agreement with classical RRKM theory. RRKM theory predicts that each product vibrational

state has an equal probability of being formed and, thus, the unimolecular rate constant is the same for forming each product vibrational state.⁵³ This is illustrated in Figure 5b where it is shown that the CH_3 vibrational energy is independent of the $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ dissociation time. Thus, discarding trajectories without product ZPE does not alter the classical $N(t)/N(0)$ plot.

V. SUMMARY

If the classical intramolecular dynamics of a decomposing molecule is ergodic on the time scale of the unimolecular reaction, an initial microcanonical ensemble of states for the molecule will decompose exponentially as assumed by RRKM theory, and with the classical microcanonical rate constant. One reason the rate constant is not the quantum microcanonical (i.e., RRKM) value is because trajectories may cross the TS without ZPE in the vibrational modes orthogonal to the reaction coordinate. For unimolecular reactions without a saddlepoint, as for $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ studied here, the trajectories allow formation of products without ZPE. For the ZPE constraint model used here, a $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ dissociating trajectory, which forms CH_3 without ZPE, has its atomic velocities reversed and “sent back” to the phase space for the vibrationally excited CH_4^* . This procedure is continued until the trajectory dissociates with ZPE in the reaction products. The total lifetime for the trajectory is the time it takes the trajectory to dissociate correctly and includes all the “returns” to the CH_4^* phase space. For the $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ dissociation simulations performed here, the population of CH_4^* molecule versus time, $N(t)$, decays exponentially as assumed by RRKM theory and with a rate constant smaller than the classical RRKM rate constant and within 40% of the harmonic quantum RRKM rate constant. In contrast, a model that discards trajectories without ZPE in the reaction products^{17–21} gives a $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ rate constant that agrees with the classical and not quantum RRKM value.

The applicability of the ZPE constraint model described here will depend on the nature of the atomic motions for the classical trajectories. For direct bimolecular reactions, without a prereaction complex, regular classical motion is expected and reversing the atomic velocities when the products are reached should simply send the trajectory back to the reactants. However, much different dynamics are expected for a unimolecular reactant such as CH_4^* studied here, whose atomic motions are irregular and chaotic, resulting in ergodicity as required for RRKM theory.⁵³ To follow the motion of a chaotic trajectory for any period of time requires very high precision in integrating the classical equations of motion. For numerical integrations on a computer, it is impossible to retain complete precision and as a result of small perturbations in the atomic positions and velocities a chaotic trajectory will “move” to another chaotic trajectory with a different unimolecular lifetime and product energies. Thus, the ZPE constraint model described here may be quite useful for studying unimolecular reactions with chaotic intramolecular dynamics.

Symplectic numerical integrators are widely used to integrate the classical equations of motion.⁵⁴ They preserve the classical phase space structure and volume but do not exactly integrate the classical equations of motion.⁵⁴ The impossibility of an exact numerical integration of the classical equations for chaotic trajectories was recently discussed within the context of implementing the semiclassical initial value representation (SC-IVR) theory.⁵⁵ For C_2H_5 dissociation, which has chaotic,

statistical unimolecular dynamics, obtaining an accurate $N(t)/N(0)$ distribution requires energy conservation, but not exact integration of the classical equations of motion.⁹

For a bimolecular reaction with a prereaction complex (or complexes), reversing the trajectories atomic velocities at a TS or at the products will not necessarily send the trajectories back to reactants. If there is chaotic dynamics in the strongly coupled, prereaction complex region of the PES, the trajectory may follow different dynamics when the velocities are reversed. It may follow the same reaction pathway but form products with ZPE. In addition, it may follow another pathway if any are available from the prereaction region; i.e., for the $\text{OH}^-(\text{H}_2\text{O}) + \text{CH}_3\text{I}$ reaction there are multiple pathways from the prereaction region.^{23,24} The ZPE constraint model described here will give trajectories for bimolecular reactions that have prereaction complexes with chaotic dynamics, multiple opportunities to dissociate correctly.

Many trajectory simulations are performed by Born–Oppenheimer direct dynamics,⁵⁴ and it is important to point out perturbations introduced into trajectories during these simulations. At each numerical integration step an electronic structure theory method is used to “directly” calculate the potential energy, gradient, and possibly Hessian for the trajectory. The electronic structure calculation is not “exactly” converged at each of these steps introducing perturbations into the trajectory dynamics. This is not expected to affect the dynamics for regular trajectories⁹ but will enhance “transitions” between chaotic trajectories.

The purely classical simulations of CH_4^* dissociation indicates that an anharmonic correction is important for the classical RRKM rate constant, i.e., $k_{\text{anh}}(E) = f_{\text{anh}}(E) k_{\text{h}}(E)$, where $f_{\text{anh}}(E)$ is 0.5 at the classical unimolecular threshold and decreases with increase in E . For the simulations reported here, the total CH_4^* energy is only in slight excess of the quantum dissociation threshold. A possible classical/quantum correspondence is that the classical $f_{\text{anh}}(E)$ at the classical threshold corresponds to the quantum anharmonic correction at the quantum threshold. With this model the anharmonic quantum RRKM rate constant is ~ 2 times smaller than the harmonic quantum RRKM rate constant and the rate constant obtained from the ZPE constrained trajectories. Clearly, a greater understanding is needed for the classical/quantum correspondence in anharmonic corrections for unimolecular decomposition.

For the $\text{CH}_4^* \rightarrow \text{H} + \text{CH}_3$ simulations reported here the vibrational energy of CH_4^* is 3 and 5 kcal/mol above the quantum dissociation threshold with ZPE in CH_3 . In future work it will be of interest to consider a broader range of energies for CH_4^* . Of particular interest will be decreasing the CH_4^* vibrational energy, which will increase the difference between the classical and quantum RRKM rate constants and $N(t)/N(0)$.⁹ The quantum RRKM rate constant approaches zero as the quantum threshold is approached, whereas the classical RRKM rate constant remains finite and only becomes zero at the classical threshold without ZPE. The ZPE constraint model proposed here will mimic this decrease in the quantum RRKM rate constant, and it will be important to determine the extent of agreement between the ZPE constrained and quantum RRKM rate constants.

The ZPE constraint model used here has some relationships to previous models constraining ZPE flow in classical trajectory simulations, and models and semiclassical theories for identifying trajectories with the proper quantum mechanical

conditions. Modification of the atomic velocities as done for the model proposed here, to impose a ZPE constraint, is similar to models that modify atomic velocities to restrain ZPE flow.^{14–16,56} In addition, requiring the dissociating trajectory to have ZPE in the products, as is done here, has some characteristics similar to those for the Gaussian weighing method^{57,58} in which trajectories are weighted according to the closeness of the final actions to integer values and the connection of initial and final actions in semiclassical theory.⁵⁹

In conclusion, the ZPE constraint model used here is promising for obtaining quantum RRKM unimolecular rate constants from classical trajectory simulations. However, considerably more applications and tests of the model are needed. It will be of interest to consider dissociation to two polyatomic fragments as for $C_2H_6 \rightarrow 2CH_3$ and apply both the hard and soft ZPE constraints.^{20,21} For the soft constraint the sum of the vibrational energy for both products must be equal to or in excess of their total ZPE, whereas for the hard constraint there is a ZPE constraint for each product. The soft constraint will give a larger rate constant and it will be of interest to determine which constraint gives a unimolecular rate constant in better agreement with quantum RRKM theory. The ZPE constraint used here may also be applied to a unimolecular reaction with a saddlepoint such as $C_2H_5 \rightarrow H + C_2H_4$. The constraint requires the vibrational energy at the saddlepoint to be equal to or in excess of its zero-point energy.

AUTHOR INFORMATION

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The research reported here is supported by the Robert A. Welch Foundation under Grant No. D-0005 and the Air Force Office of Scientific Research (AFOSR) under grant FA9550-12-1-0443. Support was also provided by the High Performance Computing Center (HPCC) at Texas Tech University, under the direction of Philip W. Smith. Parts of the computation are also performed on Robinson, a general computer cluster of the Department of Chemistry and Biochemistry, Texas Tech University, purchased by the NSF CRIF-MU grant CHE-0840493.

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